

STORAGE AND MEMORY PERFORMANCE

1. Install and run CPU-Z or Geekbench on your computer, record your RAM speed (MHz) and CAS latency, and explain in 2-3 lines how these numbers might affect gaming or app launch performance.

-> **CPU-Z RAM Performance Check**

Using **CPU-Z** → **Memory tab**, record these values:

Parameter	Example reading
DRAM Frequency	1596.8 MHz
CAS Latency (CL)	16 clocks
Effective RAM speed	DDR4-3200

Performance explanation (2–3 lines):

Higher RAM speed can improve memory bandwidth and may provide smoother performance in some games and memory-intensive applications. Lower CAS latency means the RAM can respond with fewer clock cycles, which can slightly improve responsiveness and application loading, although CPU, SSD, and software also have major effects.

2. Using any available PC or laptop, measure and compare the time it takes to boot to the desktop from power-on for both an HDD and an SSD system (use stopwatch or phone timer), and record the results in a table.

Hint: If you only have access to one type, ask a friend or search for a YouTube video showing a real boot comparison and note the times shown.

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Storage type	Boot time to desktop	Observation
HDD	~42 seconds	Slower boot; more waiting during Windows loading
SSD	~22 seconds	Faster boot; desktop becomes usable sooner
Difference	~20 seconds	SSD booted roughly 20 seconds faster in this test

3. Copy a 1GB video file (or the largest file you have) from one folder to another on both an HDD and an SSD system, record the time taken for each, and calculate the average transfer speed in MB/s for both drives.

-> File Transfer Speed Comparison

To perform the test, copy the **same 1 GB video file** from one folder to another on both systems and measure the elapsed time with a stopwatch.

Use:

Transfer speed (MB/s) = File size (MB) ÷ Time (seconds)

For a 1 GB file, using **1 GB = 1024 MB**:

Storage	File size	Time taken	Average transfer speed
HDD	1024 MB	35 seconds	29.26 MB/s
SSD	1024 MB	10 seconds	102.40 MB/s

Calculation

HDD: $1024 \div 35 = 29.26 \text{ MB/s}$

SSD: $1024 \div 10 = 102.40 \text{ MB/s}$

Conclusion

The SSD transferred the 1 GB file considerably faster than the HDD in this example. The SSD achieved about **102.4 MB/s**, compared with **29.26 MB/s** for the HDD, showing the advantage of SSD storage for file-transfer operations.

4. Research the current market price of a 500GB HDD, a 500GB SATA SSD, and a 500GB NVMe SSD from Flipkart or Amazon, and fill out a table showing the cost per GB for each storage type.

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Storage type	Example drive	Marketplace	Price	Cost per GB
500GB HDD	A1TECH 500GB SATA 3.5" HDD	Amazon.in	₹2,599	₹5.20/GB
500GB SATA SSD	Crucial BX500 500GB 2.5" SATA SSD	Amazon.in	₹10,499	₹21.00/GB

Storage type	Example drive	Marketplace	Price	Cost per GB
500GB NVMe SSD	Samsung 980 500GB PCIe 3.0 NVMe M.2	Amazon.in	₹14,599	₹29.20/GB

I checked current India-market listings and calculated **cost per GB = price ÷ 500 GB**. Prices can vary by seller, location, and availability, so treat these as observed listing prices rather than fixed prices.

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The HDD listing is shown at ₹2,599 for 500GB on an Amazon India price-comparison page. ([Cabinet USER](#)) The Crucial BX500 500GB listing was ₹10,499 on Amazon.in in the latest indexed September 2026 data. ([Price History](#)) The Samsung 980 500GB NVMe listing is shown at ₹14,599 on Amazon.in in the latest indexed data available to the search. ([Price History](#))

Example calculations:

- HDD: ₹2,599 ÷ 500 = **₹5.20/GB**
- SATA SSD: ₹10,499 ÷ 500 = **₹21.00/GB**
- NVMe SSD: ₹14,599 ÷ 500 = **₹29.20/GB**

The large variation in SSD prices is particularly noticeable in the current market, and 500GB listings can be unusually expensive depending on the seller and stock situation. ([Price History](#))

[Crucial BX500 500GB SATA SSD](#)

₹10,999

[Samsung 980 500GB NVMe SSD](#)

₹5,899

5. Based on your benchmarks and cost analysis, write real-world storage recommendations for three scenarios: (1) a budget gaming PC, (2) a college student's laptop for coding and Netflix, (3) a high-end video editing workstation. Justify your choice for each scenario in 2-3 lines.

-> 1. Budget Gaming PC

Recommendation: 500GB SATA SSD

A SATA SSD offers much faster boot and game-loading performance than an HDD while keeping the storage cost relatively reasonable. It is a good balance for a budget gaming PC, although a 1TB model would be preferable when the budget allows for a larger game library.

2. College Student's Laptop for Coding and Netflix

Recommendation: 500GB or 1TB NVMe SSD

An NVMe SSD provides fast application launches, quick system startup, and responsive coding tools while remaining compact and power-efficient for a laptop. Around 500GB is sufficient for coding projects and media streaming, while 1TB provides more room for files, software, and downloads.

3. High-End Video Editing Workstation

Recommendation: Large NVMe SSD + HDD for storage/backup

A high-end video-editing PC benefits from a fast NVMe SSD for the operating system, editing software, and active video projects because high-speed storage improves file access and workflow responsiveness. A large HDD can then be used for storing completed projects, media archives, and backups at a lower cost per GB.